# Macroscopic Traffic Variables

**Zhengbing He**, Ph.D.
https://www.GoTrafficGo.com

January 2, 2026

# Macroscopic Variables

Three main macroscopic variables:

- Density, $k$ (veh/km)
- Flow, $q$ (veh/h)
- Speed, $v$ (km/h)

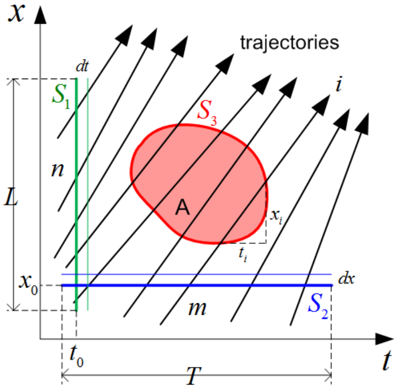
Unit is the key

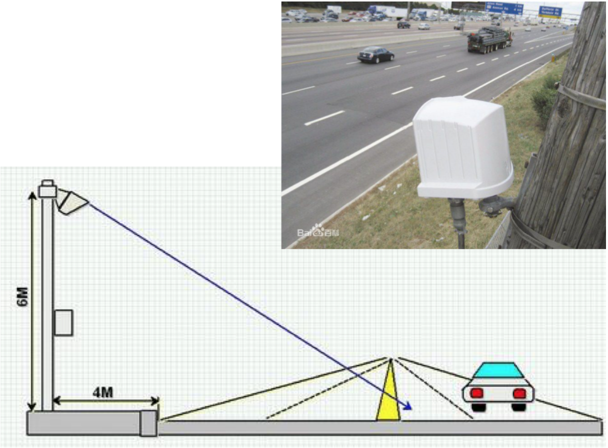
# Measurement intervals

**Three types of t-x intervals for the measurement:**

- **S1**-such as aerial photograph
- **S2**-such as loop detectors
- **S3**-such as drones.

*When trajectory data is available,* **S3** are the most appropriate, using *Edie's generalized definitions*




6M
4M

# Video surveillance

# Drone

# Germany: HighD

https://levelxdata.com/highd-dataset/

IEEE 21st International Conference on Intelligent Transportation Systems (ITSC), Maui, Hawaii, USA, November 2018

## The highD Dataset: A Drone Dataset of Naturalistic Vehicle Trajectories on German Highways for Validation of Highly Automated Driving Systems

Robert Krajewski, Julian Bock, Laurent Kloeker and Lutz Eckstein

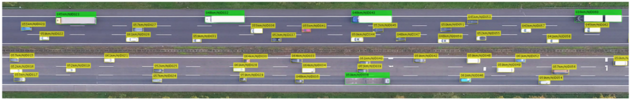
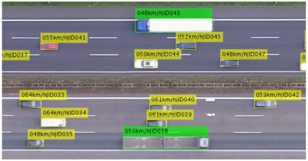
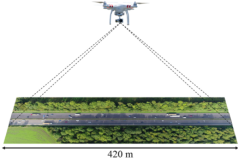


Figure 1. Example of a recorded highway including bounding boxes and labels of detected vehicles. The color of the bounding boxes indicates the class of the detected object (car: yellow, truck: green). Every vehicle is assigned a unique id for tracking and its speed is estimated over time.



# Switzerland: pNEUMA

Transportation Research Part C 111 (2020) 50–71

Contents lists available at ScienceDirect

Transportation Research Part C

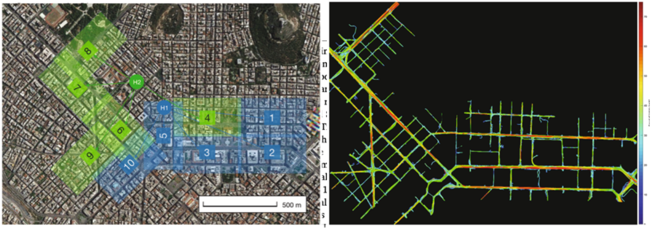
journal homepage: www.elsevier.com/locate/trc

ELSEVIER

On the new era of urban traffic monitoring with massive drone data: The *pNEUMA* large-scale field experiment

Emmanouil Barmpounakis, Nikolas Geroliminis*

*École Polytechnique Fédérale de Lausanne (EPFL), Station 18, CH-1015 Lausanne, Switzerland*



# Density, $k$ (veh/km)

**Definition**: vehicle number per distance unit at a time slice.

- **S1**: $k(x, t_0, L) = \frac{n}{L}$
  from the definition

- **S3**: $k(x, t, A) = \frac{n}{L} = \frac{ndt}{Ldt} \approx \frac{total\ \ time\ \ spent\ \ by\ \ all\ \ vehicles\ \ in\ \ A}{Area\ \ A}$
  we write as $\boldsymbol{k(A) = \frac{t(A)}{|A|}}$
  expand the definition to an area

- **S2**: $k(x_0, t, T) = \frac{\sum_m \frac{dx}{v_i}}{Tdx} = \frac{\sum_m \frac{1}{v_i}}{T}$
  apply $\frac{total\ \ time\ \ spent\ \ by\ \ all\ \ vehicles\ \ in\ \ A}{Area\ \ A}$ in S3 to S2 area

# Flow (rate), $q$ (veh/h)

**Definition**: vehicle number passing a cross-section per time unit.
We call the *maximum possible flow* as **capacity**.
Capacity of a highway lies between 1800 $\sim$ 2400 veh/h per lane.

- **S2**: $q(x_0, t, T) = \frac{m}{T}$

  from the definition

- **S3**: $q(x, t, A) = \frac{m}{T} = \frac{mdx}{Tdx} \approx \frac{Total\ \ distance\ \ covered\ \ by\ \ vehicles\ \ in\ \ A}{Area\ \ A}$

  we write as $\boldsymbol{q(A) = \frac{d(A)}{|A|}}$

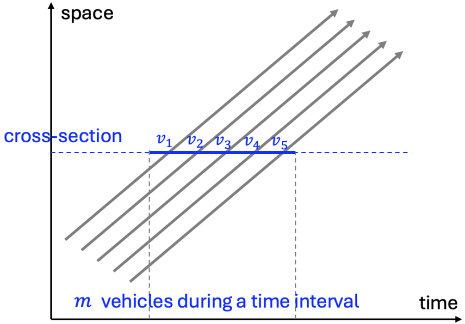
  expand the definition to an area

- **S1**: $q(x, t_0, L) = \frac{\sum_n v_i dt}{Ldt} = \frac{\sum_n v_i}{L}$

  apply $\frac{total\ \ distance\ \ covered\ \ by\ \ vehicles\ \ in\ \ A}{Area\ \ A}$ in S3 to S1 area

Time-mean Speed: averages over time at a point.

*Average speed of all vehicles passing a cross-section during a time interval.*

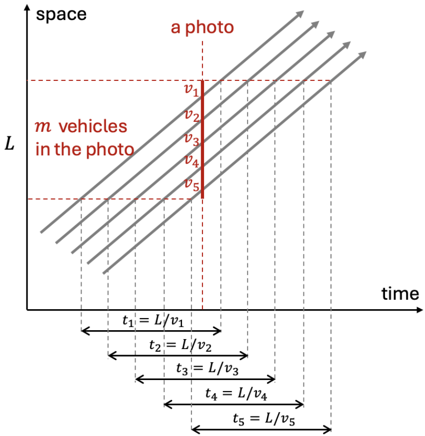



**time-mean speed**

$$v_t = \sum_{i=1}^{m} v_i$$

## Space-mean Speed: averages over space along a segment

*At time $t_0$ take a snapshot of road segment and assume each vehicle traverses the entire segment at its instantaneous speed observed in that snapshot*




travel time of each vehicle

$$t_i = \frac{L}{v_i}, \qquad i = 1,2,\ldots,m$$

total travel time

$$\sum_{i=1}^{m} t_i = \sum_{i=1}^{m} \frac{L}{v_i}$$

average travel time per vehicle

$$\frac{1}{m}\sum_{i=1}^{m} \frac{L}{v_i}$$

**space-mean speed**

-- road length divided by average travel time

$$v_s = \frac{L}{\frac{1}{m}\sum_{i=1}^{m}\frac{L}{v_i}} = \frac{1}{\frac{1}{m}\sum_{i=1}^{m}\frac{1}{v_i}} = \frac{m}{\sum_{i=1}^{m}\frac{1}{v_i}}$$

# Time-mean vs. Space-mean

Essentially, the time-mean speed represents an instantaneous state of the traffic, measured at a fixed point over a period of time, whereas the space-mean speed is based on a road segment, representing an average over a length of space (or equivalently, over the travel time along that segment under steady conditions).

| | **Time-mean speed** | **Space-mean speed** |
|---|---|---|
| **Averaging dimension** | Time | Space |
| **Measurement location** | Fixed point | Road segment |
| **Mathematical averaging** | Arithmetic | Harmonic |
| **Physical meaning** | Instantaneous | Average over space or time |

# Example: Space-mean Speed vs Time-mean Speed

Consider a road segment of length $L = 1$ km with two vehicles:

- Vehicle A travels at 60 km/h, taking 1 minute to cross the segment
- Vehicle B travels at 30 km/h, taking 2 minutes to cross the segment

What are the time-mean and space-mean speeds?

---

**Time-mean speed** (arithmetic mean at a point):

$$v_t = \frac{60 + 30}{2} = 45 \text{ km/h}$$

**Space-mean speed** (harmonic mean over the segment):

$$v_s = \frac{2}{\frac{1}{60} + \frac{1}{30}} = 40 \text{ km/h}$$

# Definition

- Homogeneous traffic: the vehicles are identical;
- Stationary traffic: the traffic state (described by the macroscopic variables) does not change over time

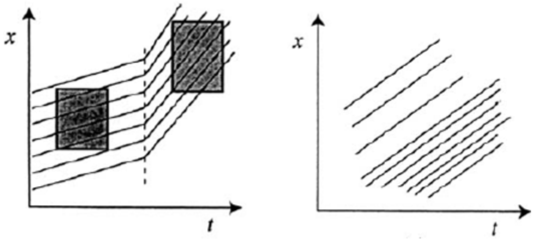



Figure: Examples of non-stationary traffic

# Fundamental Relationship

A fundamental relationship arises directly from the definitions of

- flow (vehicles per unit time),
- density (vehicles per unit distance), and
- speed

$$\boxed{q = k \cdot v}$$

QUESTION: $v$ here is time-mean speed or space-mean speed?

# Fundamental Relationship

A fundamental relationship arises directly from the definitions of

- flow (vehicles per unit time): $q = \frac{N_t}{T}$
- density (vehicles per unit distance): $k = \frac{N_s}{L}$
- speed: $v_t = \frac{1}{m}\sum_m v_i$ , $v_s = \frac{m}{\sum_m \frac{1}{v_i}}$

## Unit-based Analysis

Given

- $q \rightarrow$ veh/h
- $k \rightarrow$ veh/km
- $v_t \rightarrow$ sum($v$)/veh
- $v_s \rightarrow$ km/h

So,

- $k \cdot v_t \rightarrow$ sum($v$)/km, inconsistent with $q$ (veh/h)
- $k \cdot v_s \rightarrow$ veh/h, consistent with $q$ (veh/h)

# Fundamental Relationship

A fundamental relationship arises directly from the definitions of

- flow (vehicles per unit time): $q = \frac{N_t}{T}$
- density (vehicles per unit distance): $k = \frac{N_s}{L}$
- speed: $v_t = \frac{1}{m} \sum_m v_i$ , $v_s = \frac{m}{\sum_m \frac{1}{v_i}}$

## Derivation

Assume traffic flow is homogeneous and stationary

- During a time interval $T$, each vehicle travels an average distance of $v_s T$ (i.e., speed $\times$ time), where $v_s$ is the **space-mean speed**.
- The number of vehicles that traverse road segment during this interval:

$$N_t = k(v_s T) \quad \rightarrow \text{ density } \times \text{distance}$$

- By the definition of flow $q$ (vehicles per unit time):

$$q = \frac{N_t}{T} = \frac{k(v_s T)}{T} = k \cdot v_s$$

# Fundamental Relationship

A fundamental relationship arises directly from the definitions of

- flow (vehicles per unit time),
- density (vehicles per unit distance), and
- space-mean speed (average speed over a road segment).

$$\boxed{q = k \cdot v}$$

# Distinguishing space-mean and time-mean speed

**Example problem**: in a two-lane road, vehicles travel at 60km/h on the left lane and at 120km/h on the right lane. Assume that 1200 veh/h pass on each lane. What are the density, the flow, the (space-mean) speed and the time-mean speed on this road?

**The solution** is:

- $q = 1200 * 2 = 2400 veh/h$
- $k = 1200/60 + 1200/120 = 30 veh/km$
- $v = q/k = 80 km/h$
- $v_t = (120 + 60)/2 = 90 km/h$

# Speed, $v$ (km/h)

Assume traffic is homogeneous and stationary, and apply $q = kv$

- **S1**: using the flow and density equations in S1

$$v(x, t_0, L) = \frac{\sum_n v_i}{L} / \frac{n}{L} = \frac{1}{n} \sum_n v_i$$

**Note**: space-mean = time-mean, **under above assumption**

- **S2**: using the flow and density equations in S2

$$v(x_0, t, T) = \frac{m}{T} / \frac{\sum_m \frac{1}{v_i}}{T} = \frac{m}{\sum_m \frac{1}{v_i}}$$

- **S3**: Using definition:

$$v(x, t, A) = \frac{q(x,t,A)}{k(x,t,A)} \approx \frac{\textit{Total distance covered by vehicles in A}}{\textit{Total time spent by vehicles in A}}$$

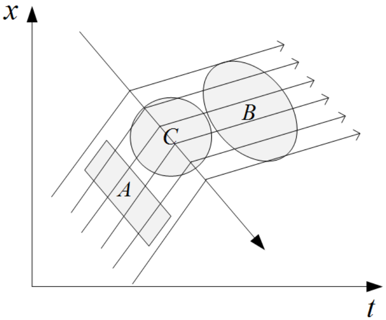
we write as $\boldsymbol{v(A) = \frac{d(A)}{t(A)}}$

the measurement area A no longer appears

# Summary about Edie's definitions

Edie's definitions (for **S3**) are important because it allows one to compare the behavior of two traffic streams even if they have been observed in different ways, such as the comparison of A and B.

One should try to avoid the non-stationary area, such as C, because one value of the macroscopic variable only corresponds to one traffic state.



# Summary about macroscopic traffic variables

The discrete nature of traffic requires time intervals of at least half a minute if we want to achieve meaningful macroscopic information. When the time intervals exceed a duration of five minutes, certain dynamic characteristics are lost.

Thank you!